AWS Solution Architect

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TECHNOLOGY

Databases

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Learning Objectives

By the end of this lesson, you will be able to:

- Create and manage a MySQL database for efficient data storage and retrieval
- Set up and utilize a DynamoDB database to handle NoSQL data and achieve scalability
- Establish DynamoDB global table replicas for data redundancy, ensuring high availability and fault tolerance
- Utilize Redshift for data warehousing and perform analytical queries to extract valuable insights from your data

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TECHNOLOGY

Introduction to Databases

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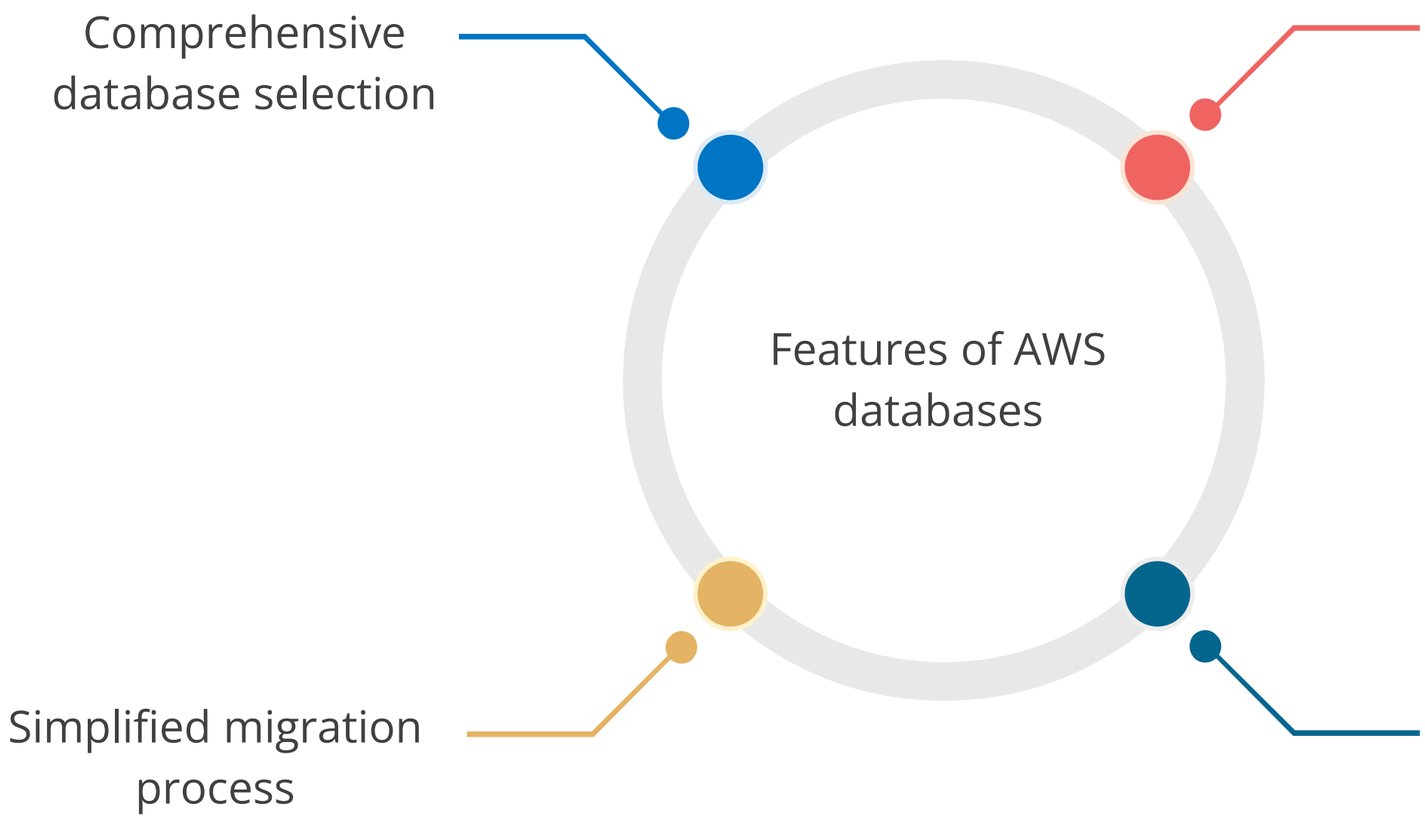
Databases: Overview

AWS provides a broad selection of purpose-built databases to help save, grow, and build a robust cloud database.



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Types of AWS Databases

AWS databases can be categorized into:

Relational

For traditional applications, ERP, CRM, and e-commerce

Key-value

For high-traffic web applications, e-commerce, and gaming applications

In-memory

For caching, session management, and geospatial applications

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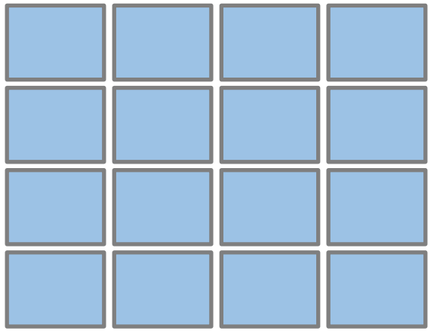
Relational Databases

It is also known as SQL databases, are the most common type of databases.

Structured data



They are used to store structured data in a set of tables that are formally described.

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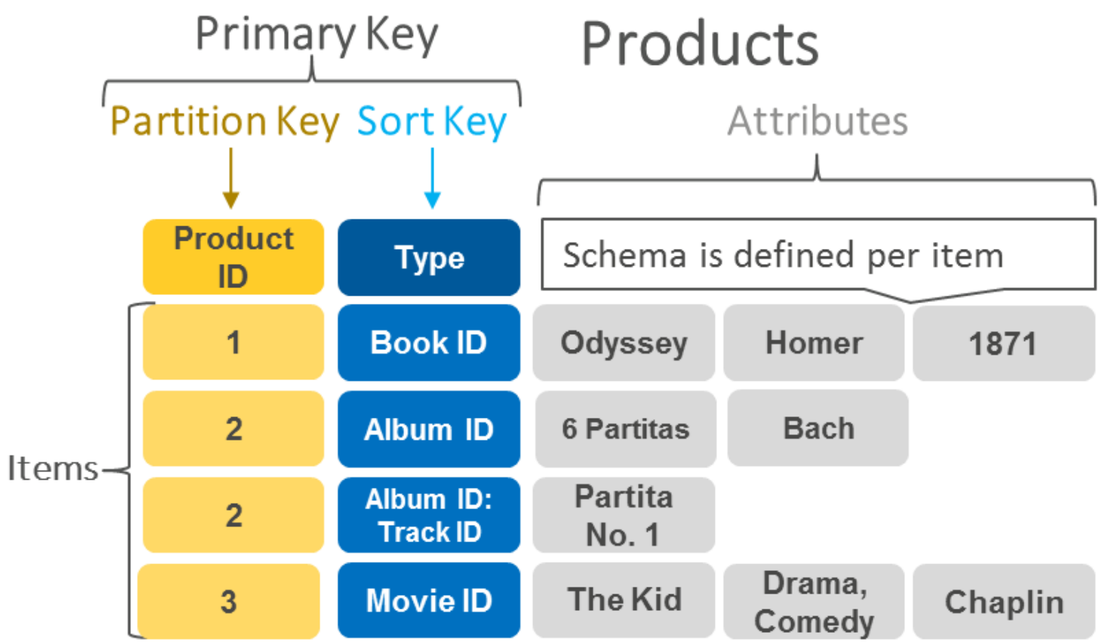
Key-Value Databases

It is a type of non-relational database that utilizes a key-value method for storing data. These databases are used in high-traffic and high-computing applications.



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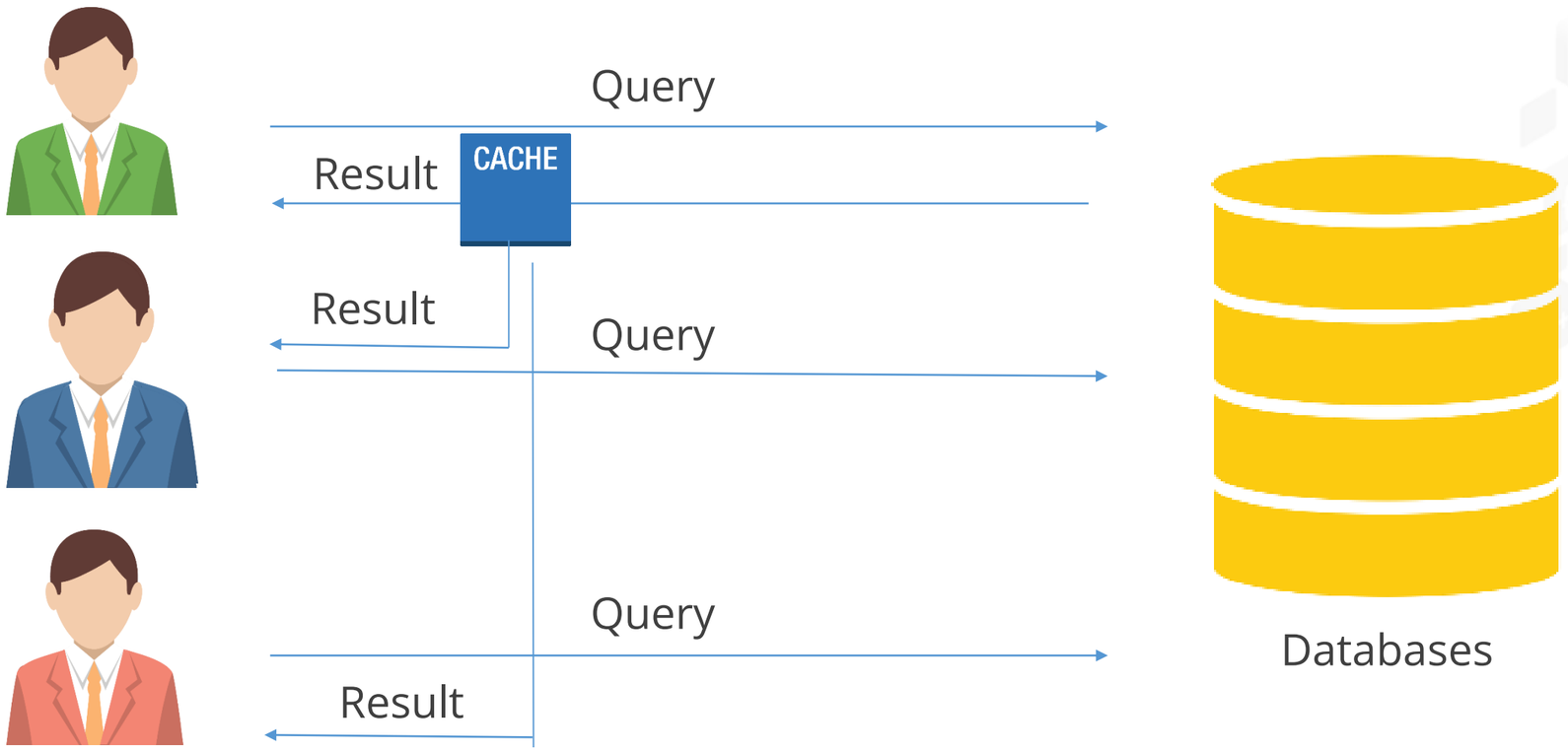
In-Memory Databases

It uses cache-based storage to reduce the database infrastructure load and enhance user response time.



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Quick Check

You need to choose an AWS database service for an application that requires a fully managed, highly available relational database with support for complex queries and transactions. Which service should you use?

A. Amazon DynamoDB

B. Amazon RDS

C. Amazon Redshift

D. Amazon Neptune

Amazon Relational Database Service (RDS)

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Amazon RDS

Amazon Relational Database Service (Amazon RDS) simplifies the process of setting up, operating, and scaling a relational database in the AWS cloud.

Here are some of the benefits of Amazon RDS:

- Ease of administration
- Scalability
- Security
- Cost-effectiveness
- Fast performance

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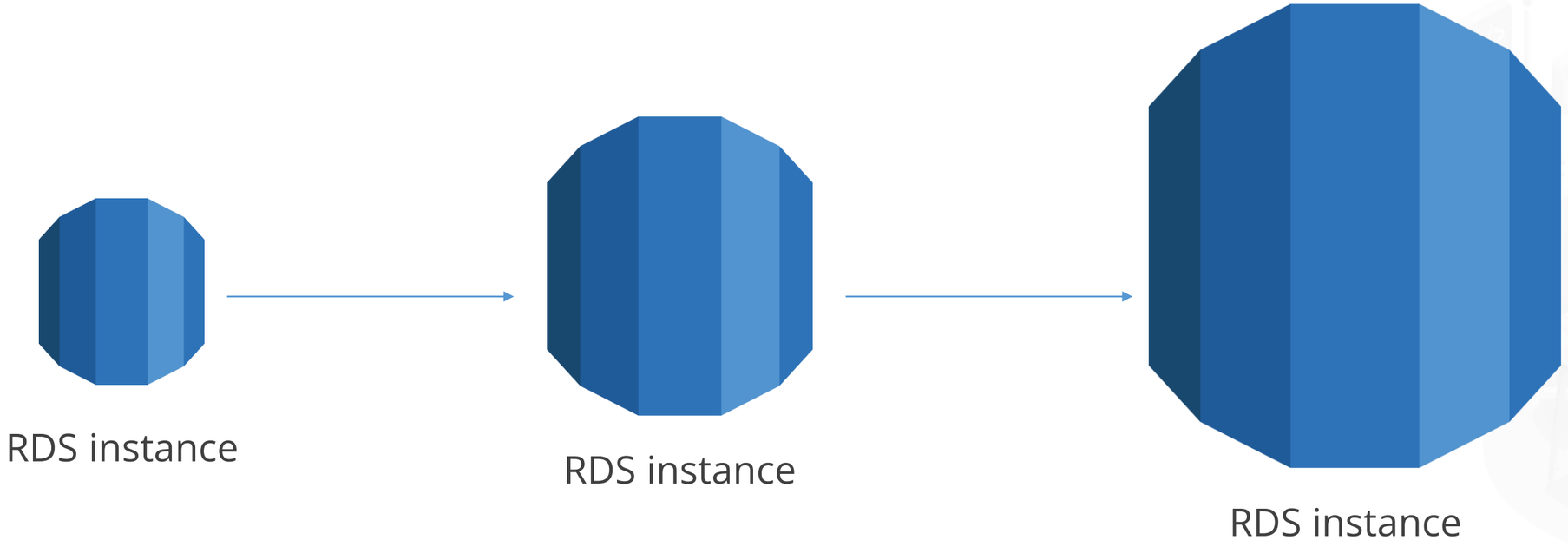
Amazon RDS

It enables users to scale the compute and memory resources of their deployments up to 32 vCPUs and 244 GiB of RAM.



Scaling operations usually finish within minutes.

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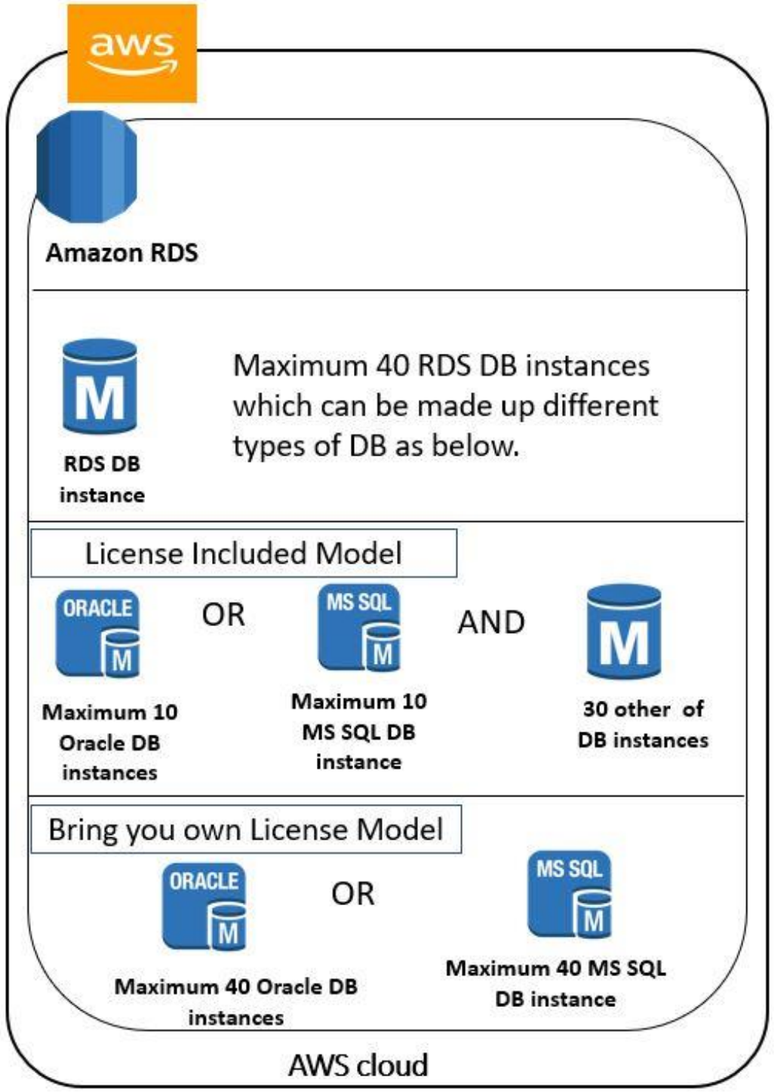
DB Instance



- A DB instance, the fundamental building block of Amazon RDS, is a standalone database environment that runs in the Cloud.
- Users can create multiple databases within a DB instance and access them using standard client tools and applications.

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RDS classes

It determines the computation and memory capacity of an Amazon RDS DB instance.

They can be categorized into three types:

- General-purpose
- Memory-optimized
- Burstable performance

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Amazon RDS Database Engines

Amazon RDS supports several database engines:

- Amazon Aurora
- PostgreSQL
- ORACLE
- MySQL
- Microsoft SQL Server
- MariaDB

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Limitations of Amazon RDS for Microsoft SQL

Certain server-level roles in Microsoft SQL are currently unavailable in Amazon RDS. These include:

- bulkadmin
- dbcreator
- diskadmin
- securityadmin
- serveradmin
- sysadmin

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Limitations of Amazon RDS for MySQL

Several features of MySQL are currently unavailable in Amazon RDS, such as:

- Authentication plugin
- Error logging to the system log
- Group replication plugin
- InnoDB tablespace encryption
- Password strength plugin
- Persisted system variables
- Replication filters
- Semisynchronous replication
- Transportable tablespace
- X Plugin

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Limitations of Amazon RDS for MySQL

Certain Oracle privileges are currently unavailable in Amazon RDS, including:

ORACLE®

- Alter database
- Alter system
- Create any directory
- Drop any directory
- Grant any privilege
- Grant any role

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Use Cases of Amazon RDS

Amazon RDS is commonly used for:

- Web and mobile applications
- E-commerce applications
- Online games

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Amazon RDS Backups

It automatically creates and securely stores backups of database instances in Amazon S3 for a specified retention period.

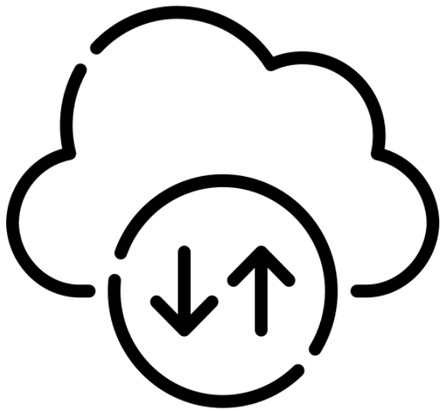
The types of Amazon RDS backups are:

Database snapshots

Point-in-time restores

Snapshot copies

Automated backups



Snapshot sharing

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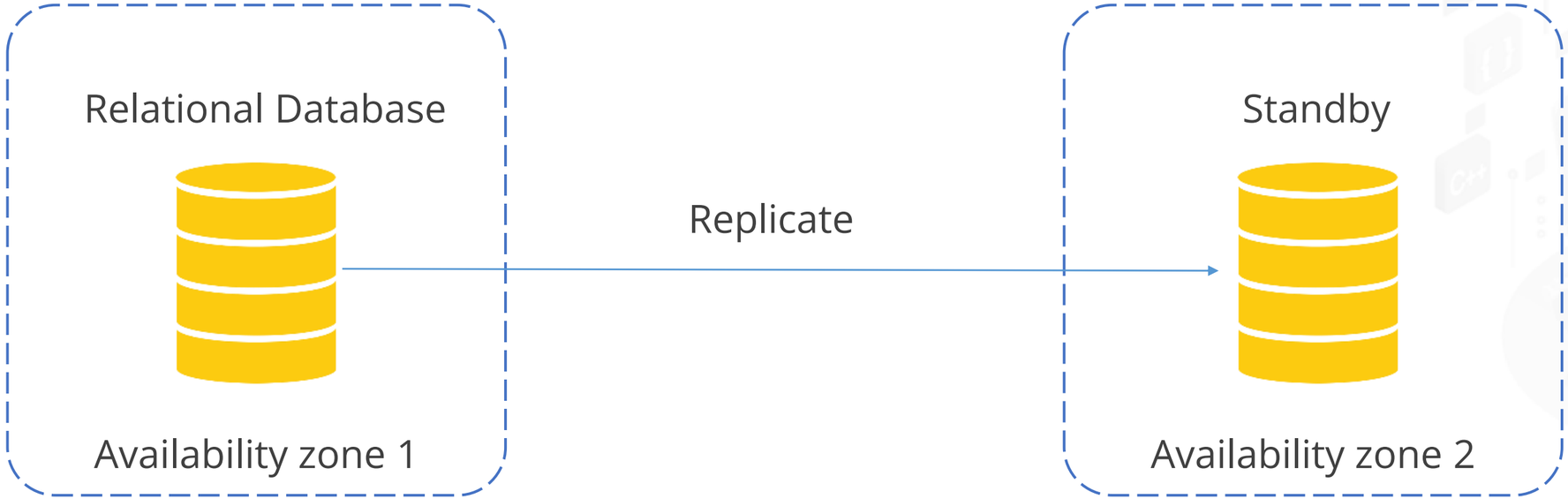
Multi-Availability Zone Deployments

It synchronously replicate data to a standby instance in a different Availability Zone.



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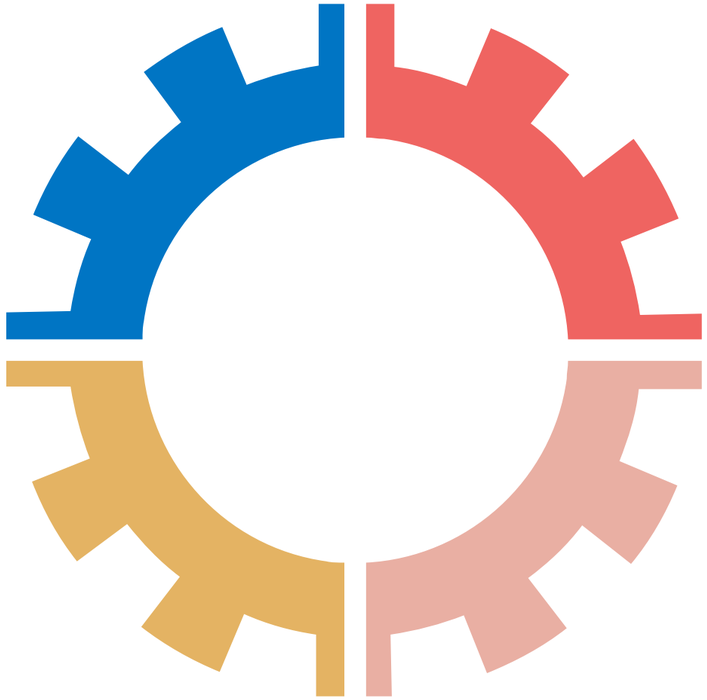
Multi-Availability Zone Deployments

Here are some benefits of multi-availability zone deployments:

Enhanced durability:
The replica of the data stays up-to-date with the primary database.

Increased availability:
If an Availability Zone fails, the backup is ready in less than 30 seconds.



Stable performance:
Multi-AZ deployments do not suspend I/O activity on your primary during backup.

Automatic failover:
Amazon RDS automatically initiates a failover to the up-to-date standby.

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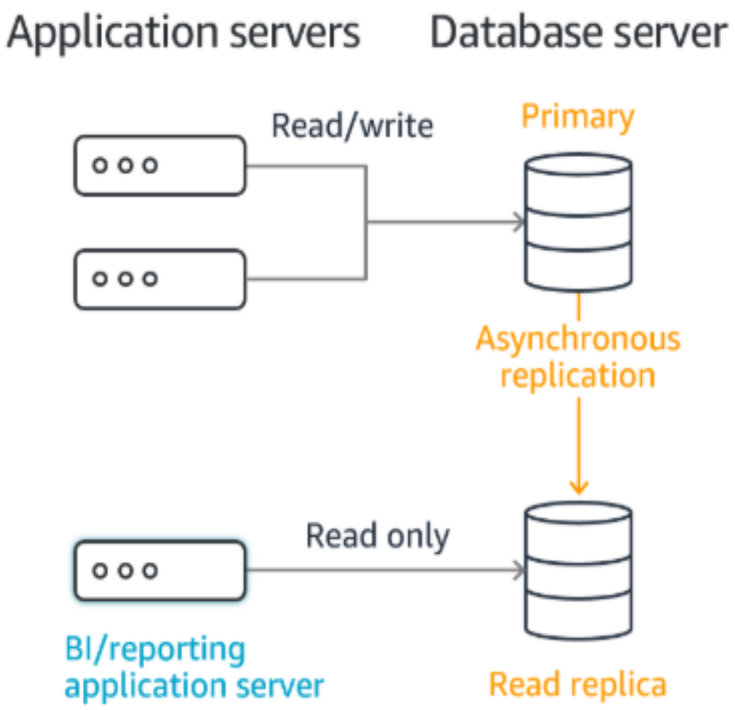
Read Replicas

Amazon RDS Read Replicas enhance the performance and durability of RDS database (DB) instances.



Here are some benefits of Read Replicas:

- Enhanced performance
- Increased availability
- Designed for security

Source: https://aws.amazon.com/rds/features/read-replicas/

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Amazon Aurora

It is a relational database that combines the speed and availability of high-end commercial databases with the simplicity and cost-effectiveness of open-source databases.

The benefits of Amazon Aurora include:

- High performance and scalability
- High availability and durability
- Robust security
- Compatibility with MySQL and PostgreSQL
- Full manageability
- Support for migration

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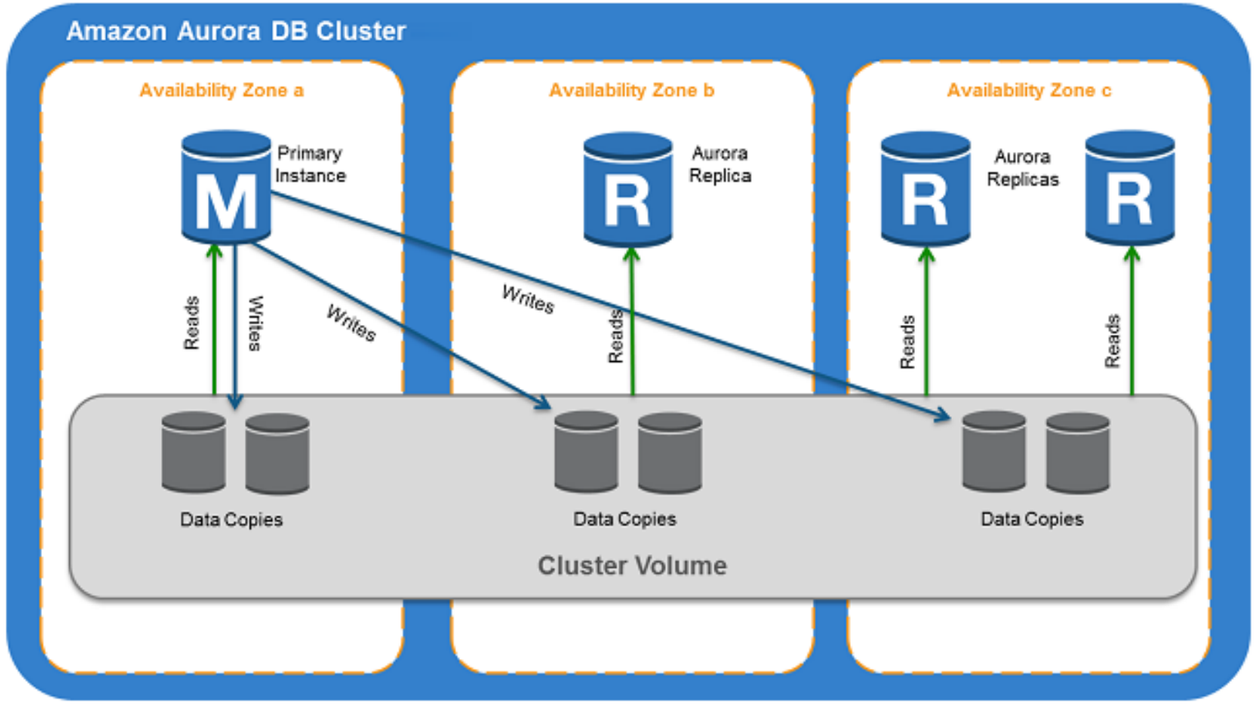
Amazon Aurora DB Cluster

It includes one or more DB instances and a cluster volume that manages the data for those DB instances.



Source: https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Overview.html

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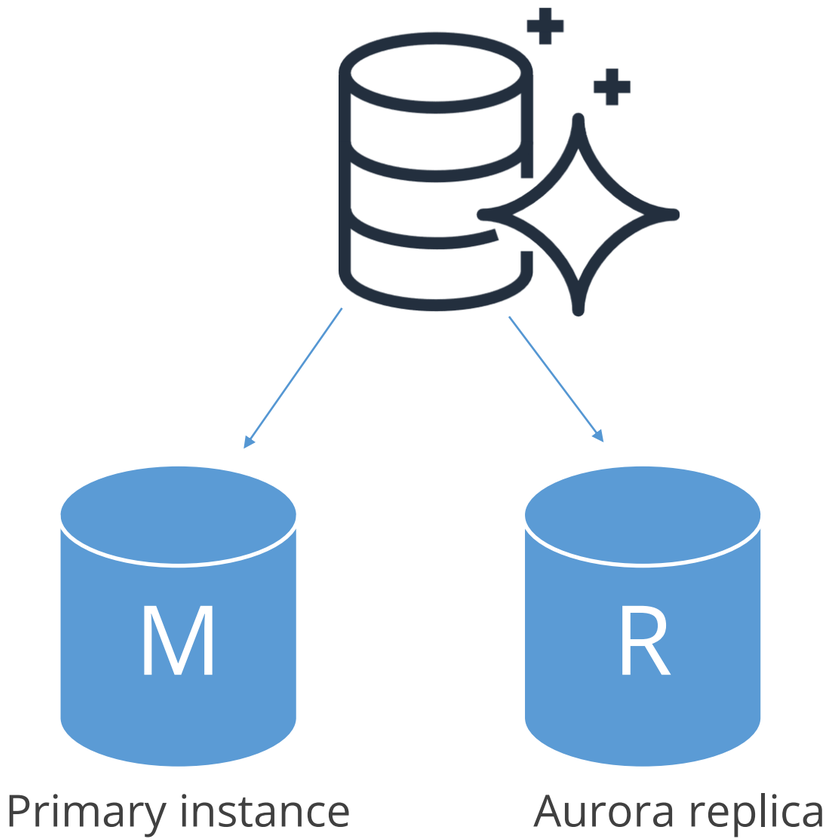
Amazon Aurora DB Cluster

An Aurora DB Cluster in Amazon consists of two types:



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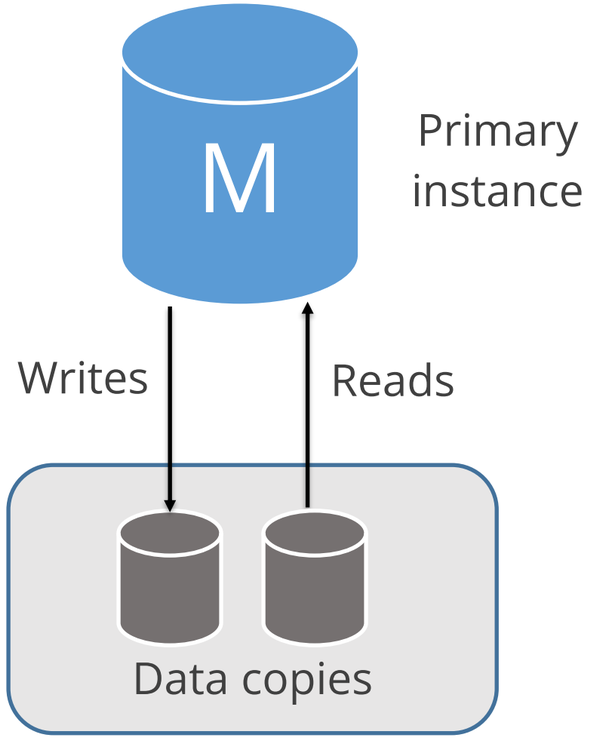
Amazon Aurora DB Cluster: Primary Instance

In each Aurora DB cluster, the primary instance handles both read and write operations, updating the data in the cluster volume with every alteration.



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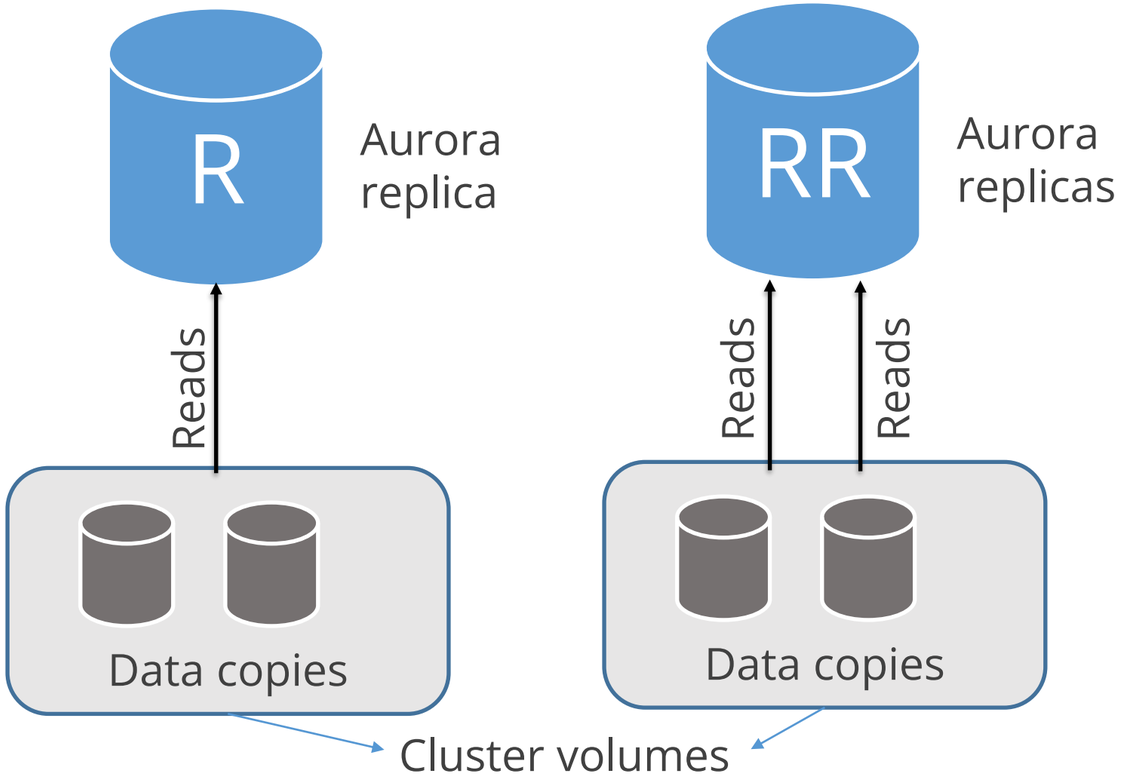
Amazon Aurora DB Cluster: Aurora replica

An Aurora replica supports only read operations and connects to the same storage disk as the primary DB instance.



Each Aurora DB cluster can contain up to 15 Aurora replicas in addition to the primary DB instance.

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Amazon Aurora Serverless

It is a configuration that automatically scales. It starts, shuts down, and adjusts capacity based on the application's needs.

There are two versions of Amazon Aurora Serverless:

- Amazon Aurora Serverless v1
- Amazon Aurora Serverless v2

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Amazon Aurora Serverless v1

It is a configuration that adjusts capacity dynamically without manual intervention.

- It allows the computational capacity of a DB cluster to be scaled up or down according to the application's needs.
- Unlike other versions, the capacity of DB clusters created by Aurora Serverless v1 must be managed manually.

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Amazon Aurora Serverless v1

The use cases for Amazon Aurora Serverless v1 include:

- Infrequently-used applications
- Development and test databases

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Amazon Aurora Serverless v1

Here are some benefits of Aurora Serverless v1:

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Amazon Aurora Serverless v2

It helps users to automate the procedures for workload monitoring and database capacity adjustment.

Here are some benefits of Aurora Serverless v2:

- Highly scalable
- Simple
- Highly available
- Transparent
- Cost-effective
- Durable

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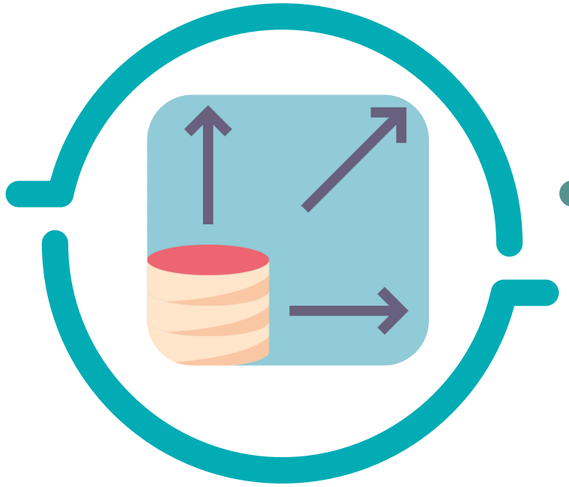
Amazon Aurora Serverless v2

Use cases for Amazon Aurora Serverless v2 include:

Managing enterprise database fleets

Supporting software as a service applications



Distributing scaled-out databases across multiple servers

Handling unpredictable workloads

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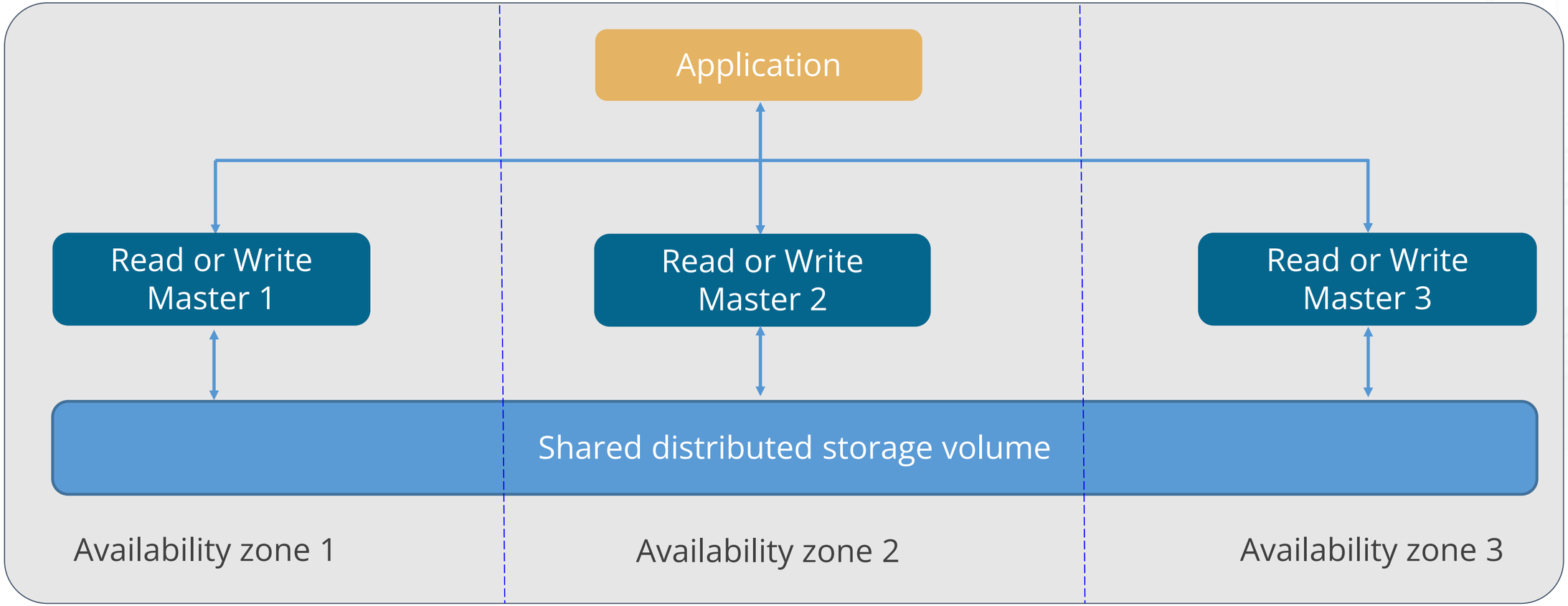
Aurora Multi-Master

It allows users to create multiple read and write master instances across several availability zones.



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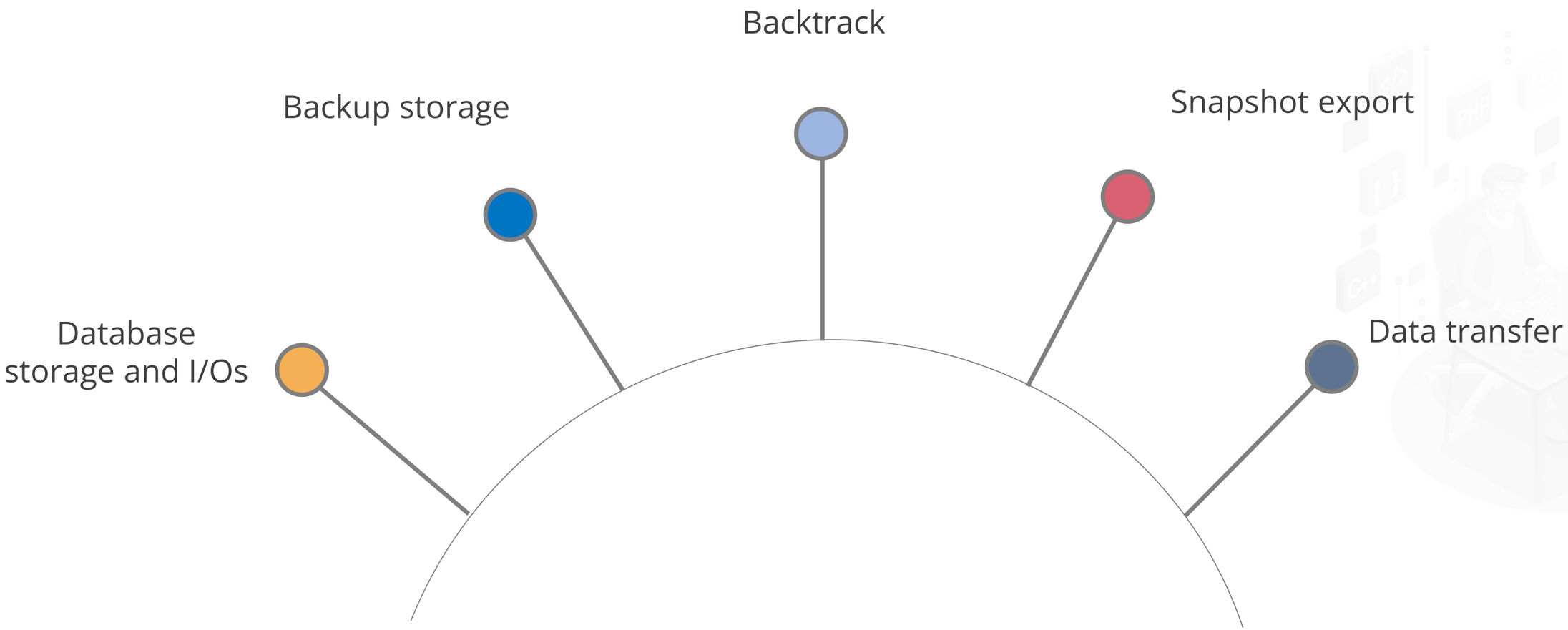
Amazon Aurora Costs

Costs associated with Amazon Aurora include:



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Quick Check

You are developing an application that requires a highly available relational database with automated backups and the ability to scale compute and storage independently. Which Amazon RDS feature should you use to achieve this?

A. Multi-AZ Deployment

B. Read Replicas

C. Amazon Aurora

D. Database Snapshots

DynamoDB

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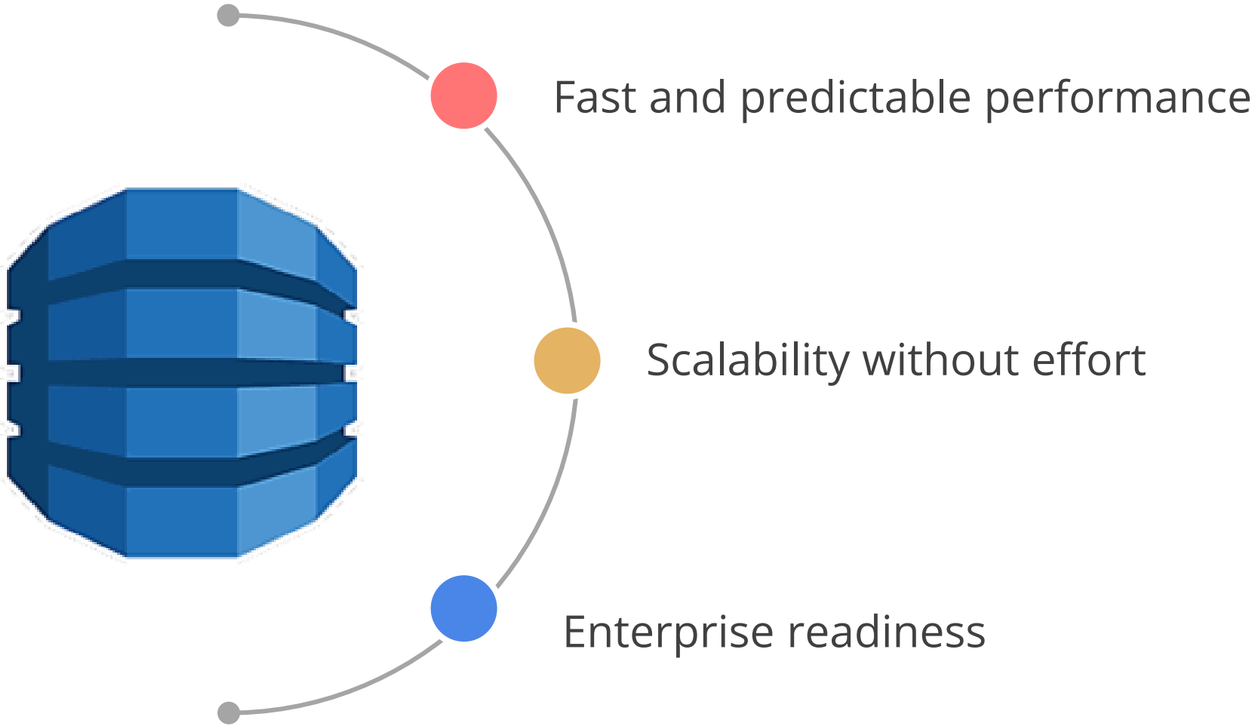
Amazon DynamoDB

It is a fully managed NoSQL database service. It offers:



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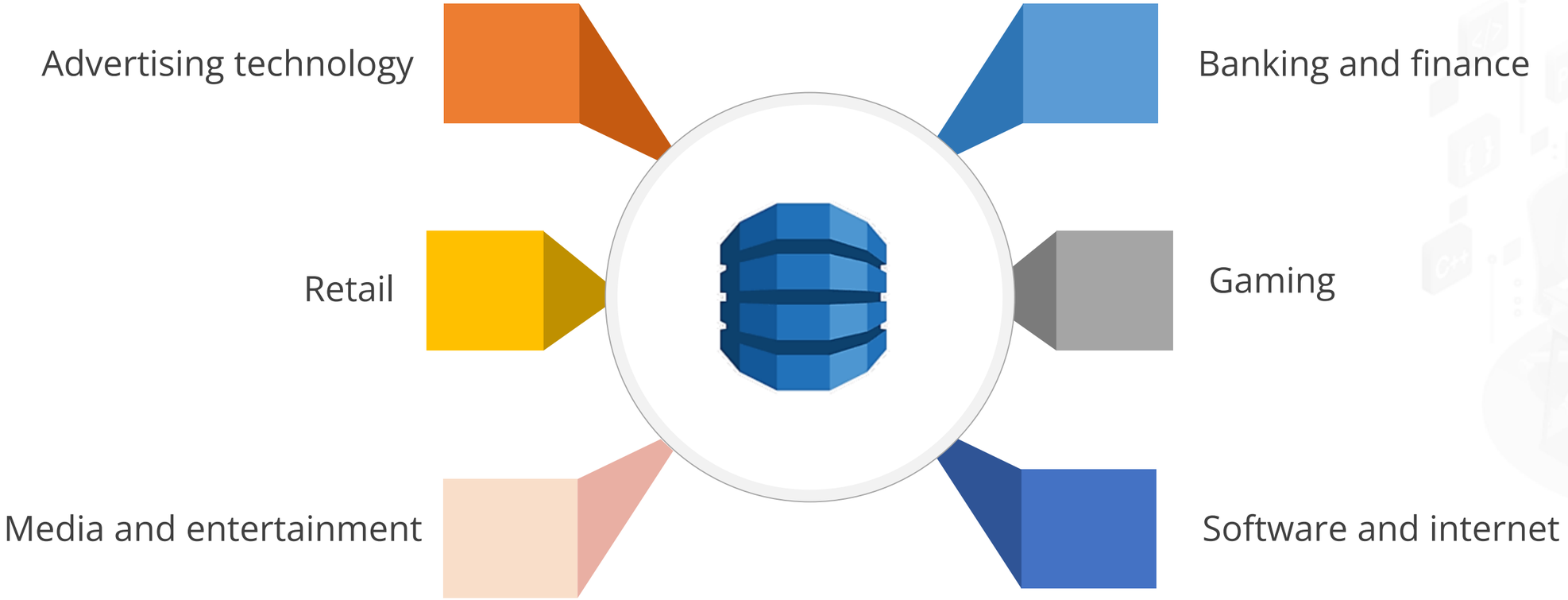
Amazon DynamoDB: Use Cases

Amazon DynamoDB is widely used in:



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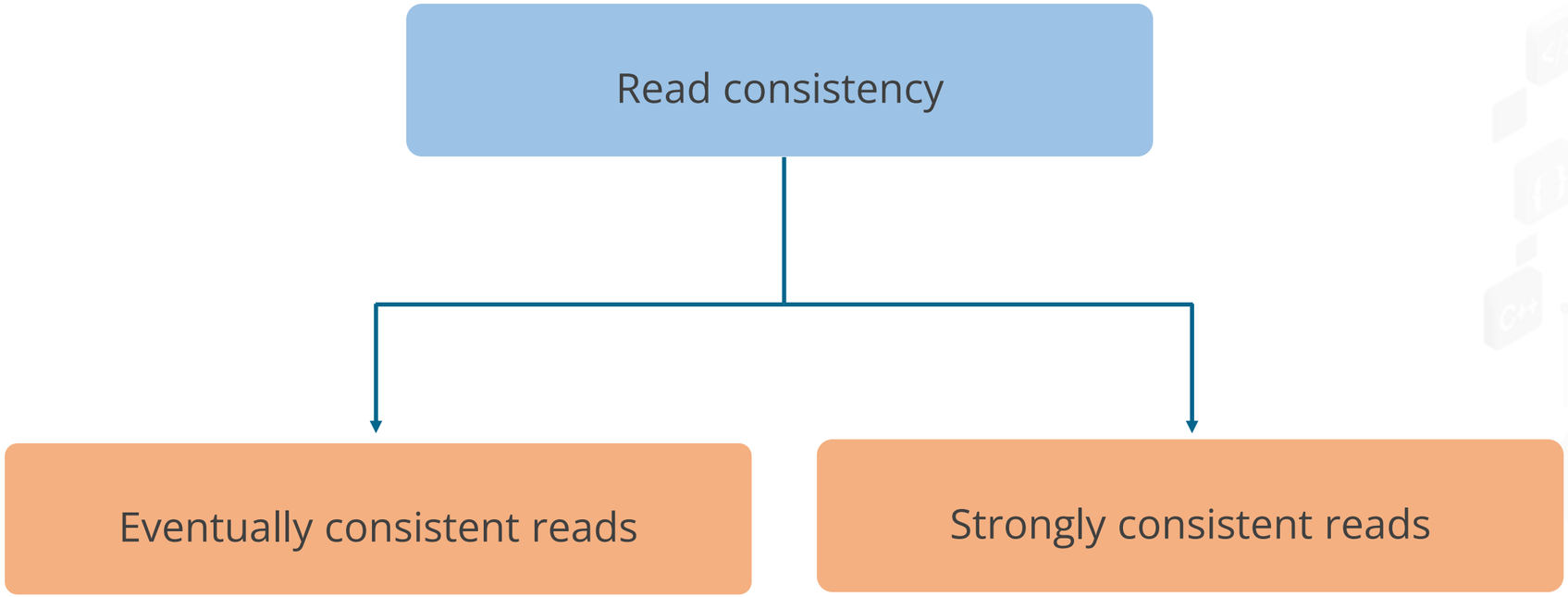
Amazon DynamoDB: Read Consistency

Amazon DynamoDB provides two types of read consistency:



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Amazon DynamoDB Global Tables

- Global tables offer a fully managed, multi-region, and multi-master database.
- Amazon ensures fast, local read-and-write performance.
- Global tables automatically replicate DynamoDB tables across the chosen AWS Regions.

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Amazon DynamoDB: Reading an Item

To read an item from a DynamoDB table, use the **get-item** command

An example of a read operation:

```
aws dynamodb get-item \
  --table-name Catalog \
 --key '{"Id":{"N":"1"}}'
```

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Amazon DynamoDB

To write an item in a DynamoDB table, users can use the following commands:

PutItem

This command creates a new item. If an item with the same key already exists, the new item replaces it.

DeleteItem

This command deletes the item associated with the specified key.

UpdateItem

If the specified key doesn't exist, this command adds a new item. If the key exists, it updates the item's attributes.

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Amazon DynamoDB

In DynamoDB, a global secondary index (GSI) has a partition key and a sort key distinct from the base database's main key.

This index allows queries to access data from multiple base table partitions, making it global.

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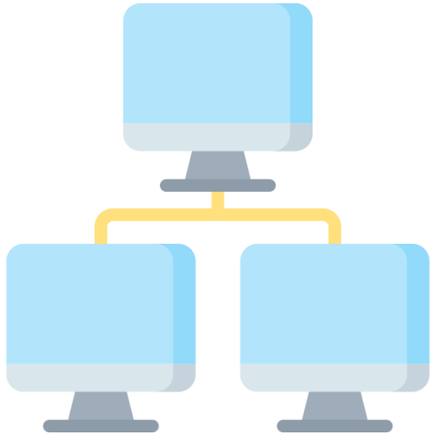
Amazon DynamoDB

In DynamoDB, a local secondary index allows for flexible queries within a specific partition.



Unlike the global secondary index (GSI), which can span multiple partitions, the LSI operates within the constraints of a specific partition key.

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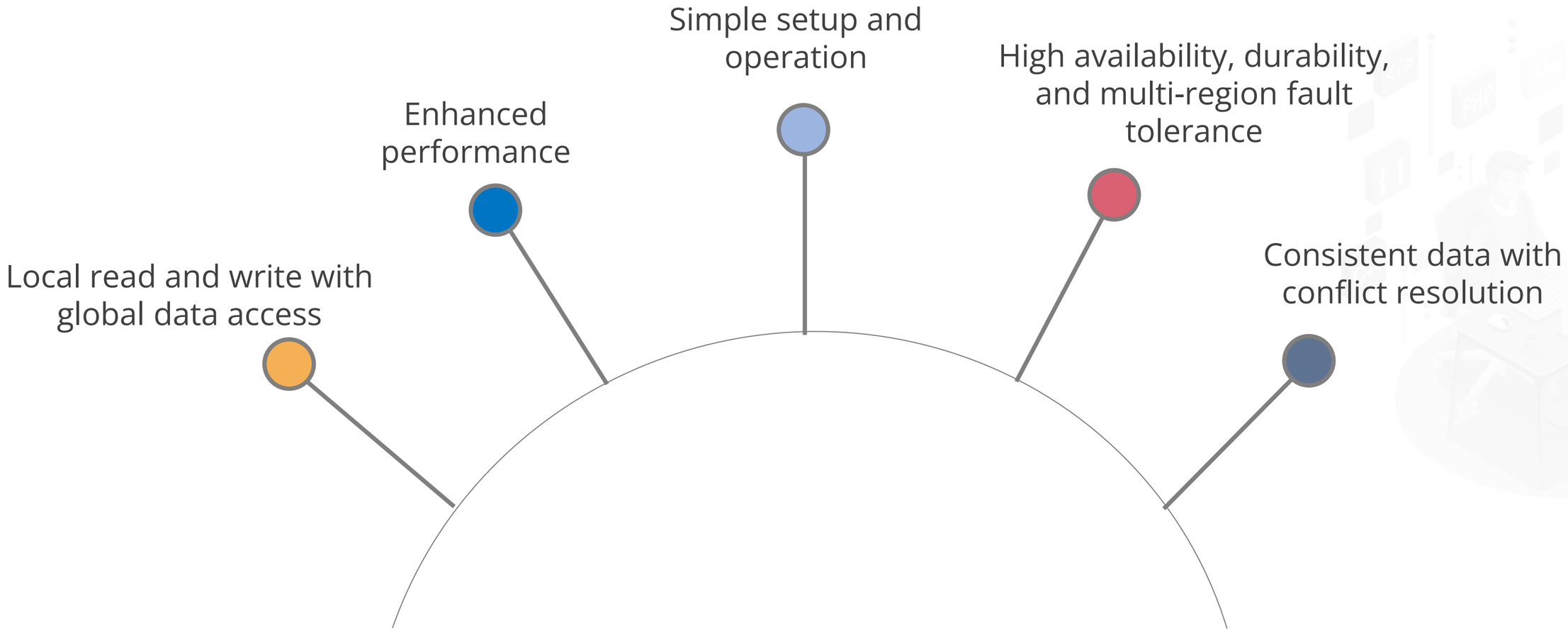
DynamoDB Global Tables

The benefits of Amazon DynamoDB global tables are as follows:



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Case Study: Duolingo

Duolingo utilizes Amazon DynamoDB to store 31 billion items, supporting its online learning platform that offers lessons in 80 languages.

- 31 billion items
- 80 different languages
- 24,000 read units per second and 3300 write units per second

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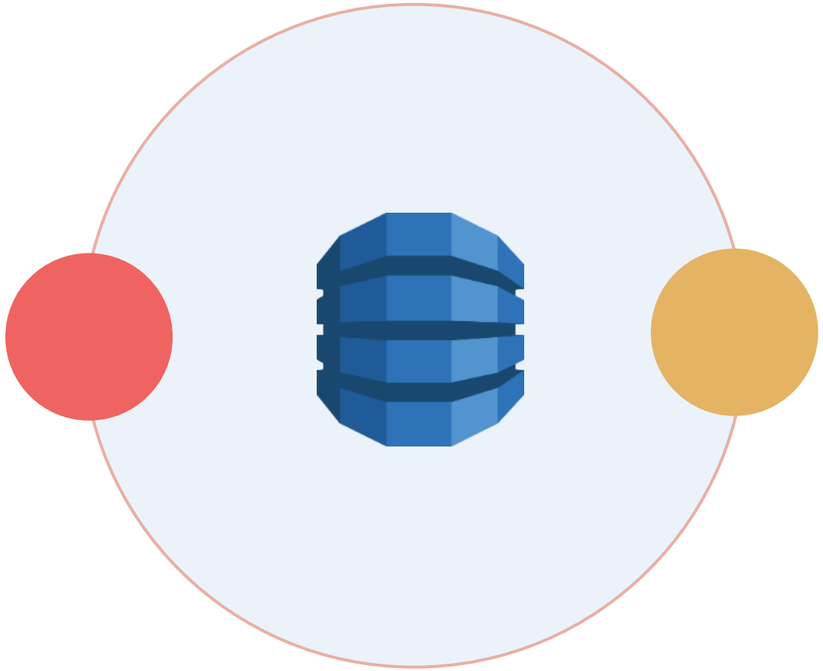
Amazon DynamoDB: Costs

Amazon DynamoDB offers the following cost structures:

On-demand capacity mode



Provisioned capacity mode

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Amazon DynamoDB: Features

Read consistency

- DynamoDB supports both eventually consistent and strongly consistent reads.
- Reading data from a DynamoDB table may not immediately reflect the results of a recently completed write operation.
- DynamoDB maximizes read throughput.

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Amazon DynamoDB: Features

Capacity management

- DynamoDB manages throughput capacity for read and write operations.
- It can scale up or down to accommodate variable read/write demands.
- Developers can select the capacity mode that best aligns with their application's requirements.

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Amazon Keyspaces for Apache Cassandra

You can run Cassandra workloads on AWS using the same Cassandra application code and developer tools.

- Apache Cassandra is scalable and highly available.
- The Apache Cassandra-compatible database service is now available for preview in 18 AWS regions.

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Amazon Keyspaces for Apache Cassandra: Benefits

Here are some benefits of Apache Cassandra:

01 Offers high performance and scalability

02 Provides high availability

03 Ensures high security

04 Remains compatible with Apache Cassandra

05 Requires no servers to manage

06 Maintains performance at scale

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Amazon Keyspaces for Apache Cassandra: Use Cases

The use cases for Apache Cassandra are:

- Building low-latency applications
- Migrating existing Cassandra workloads to the cloud
- Developing applications using open-source technologies
- Serving as a data store for applications

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Amazon DocumentDB

It provides powerful and intuitive APIs for flexible and agile development.

- The data is represented as a JSON document.
- The document data model is intuitive for developers.
- The data is persisted in the database using the same document model format used in the application code.

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Amazon DocumentDB: Benefits

Amazon DocumentDB, with its MongoDB compatibility, offers several benefits such as:

Reliability:
Amazon DocumentDB provides dependable database service with minimal downtime.

Speed:
It is fast, facilitating efficient and prompt data operations.

Management:
It is a fully managed service, removing the need for hands-on database management.

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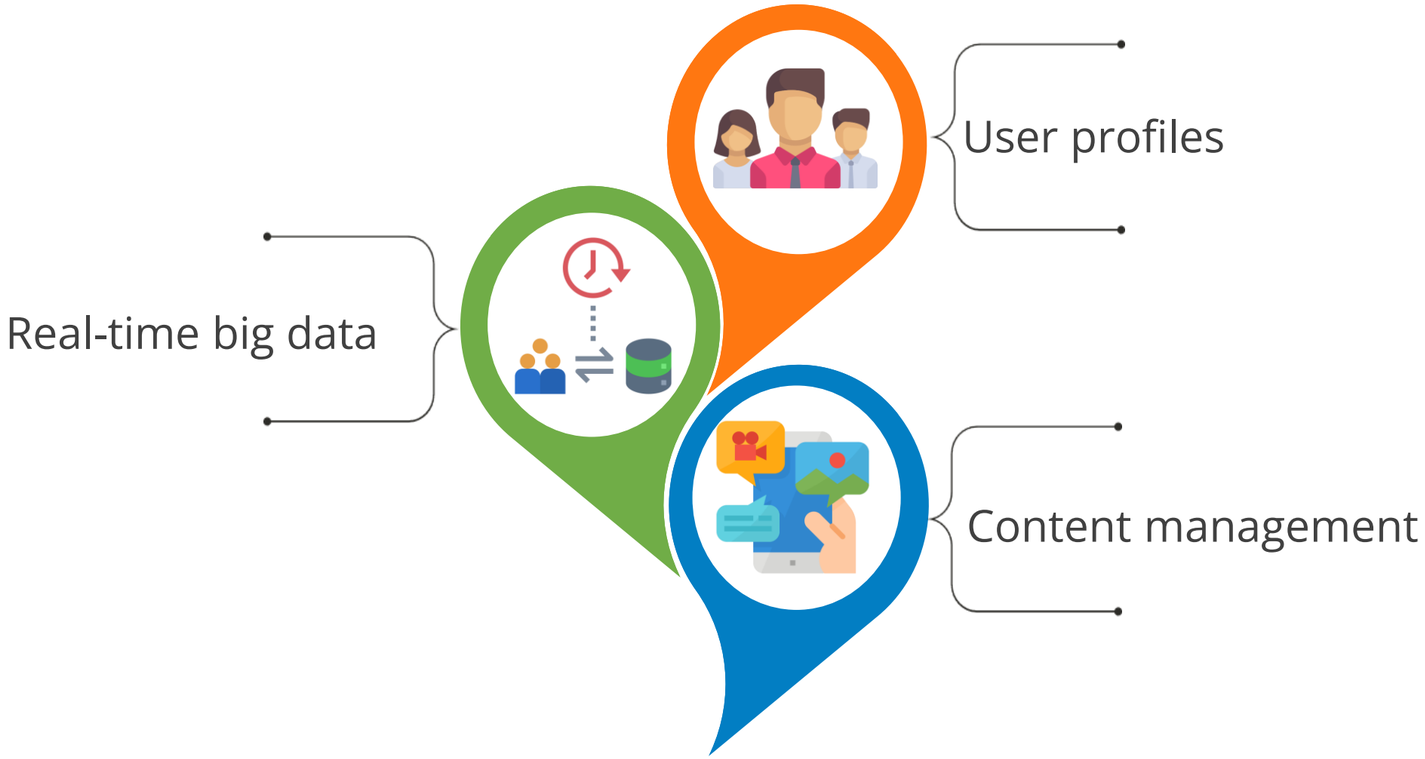
Amazon DocumentDB

Amazon DocumentDB has several use cases, including:



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Implementing Database Operations in DynamoDB

Duration:15 min

Problem Statement:

You have been assigned a task to create a database table using DynamoDB to store and read items in a database.

Outcome:

You will be able to create a database table that can store and read items using the DynamoDB console.

Note: Refer to the demo document for detailed steps:
01_Implementing_Database_Operations_in_DynamoDB

ASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create a table
2. Store and read the items

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Creating a Replica of DynamoDB

Duration:15 min

Problem Statement:

You have been assigned a task to create a global table replica and make data available across different regions using DynamoDB for improving data accessibility and enhancing application performance in geographically dispersed environments.

Outcome:

You will be able to set up a global table replica and make data available across different regions using DynamoDB.

Note: Refer to the demo document for detailed steps:
02_Creating_a_Replica_of_DynamoDB

ASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create a replica in DynamoDB

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Quick Check

You are building an application that requires a NoSQL database with single-digit millisecond latency at any scale. The database must be fully managed and support key-value and document data structures. Which AWS service should you choose?

A. Amazon RDS

B. Amazon Redshift

C. Amazon DynamoDB

D. Amazon ElastiCache

Amazon ElastiCache

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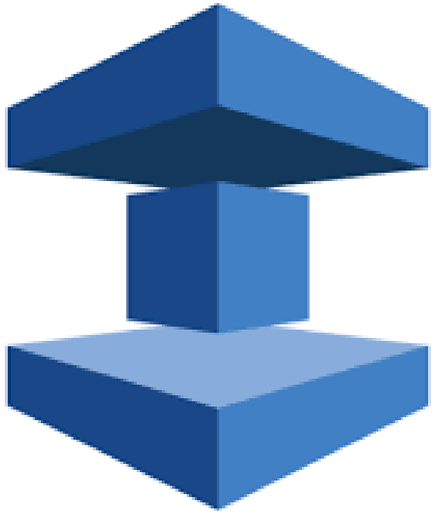
Amazon ElastiCache

It is a web service that simplifies the deployment, operation, and scaling of an in-memory data store or cache in the cloud.



Key features of Amazon ElastiCache include:

- Easy management through the AWS management console
- Engine protocol compatibility
- Detailed engine node monitoring at no additional cost via Amazon CloudWatch
- Pay-per-use model for resource consumption

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Amazon ElastiCache: Benefits

Here are some benefits of Amazon ElastiCache:

- Fast performance
- Scalable
- Fully manageable

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Amazon ElastiCache: Use Cases

Some of the use cases of Amazon ElastiCache:

Caching

Geospatial

Media streaming

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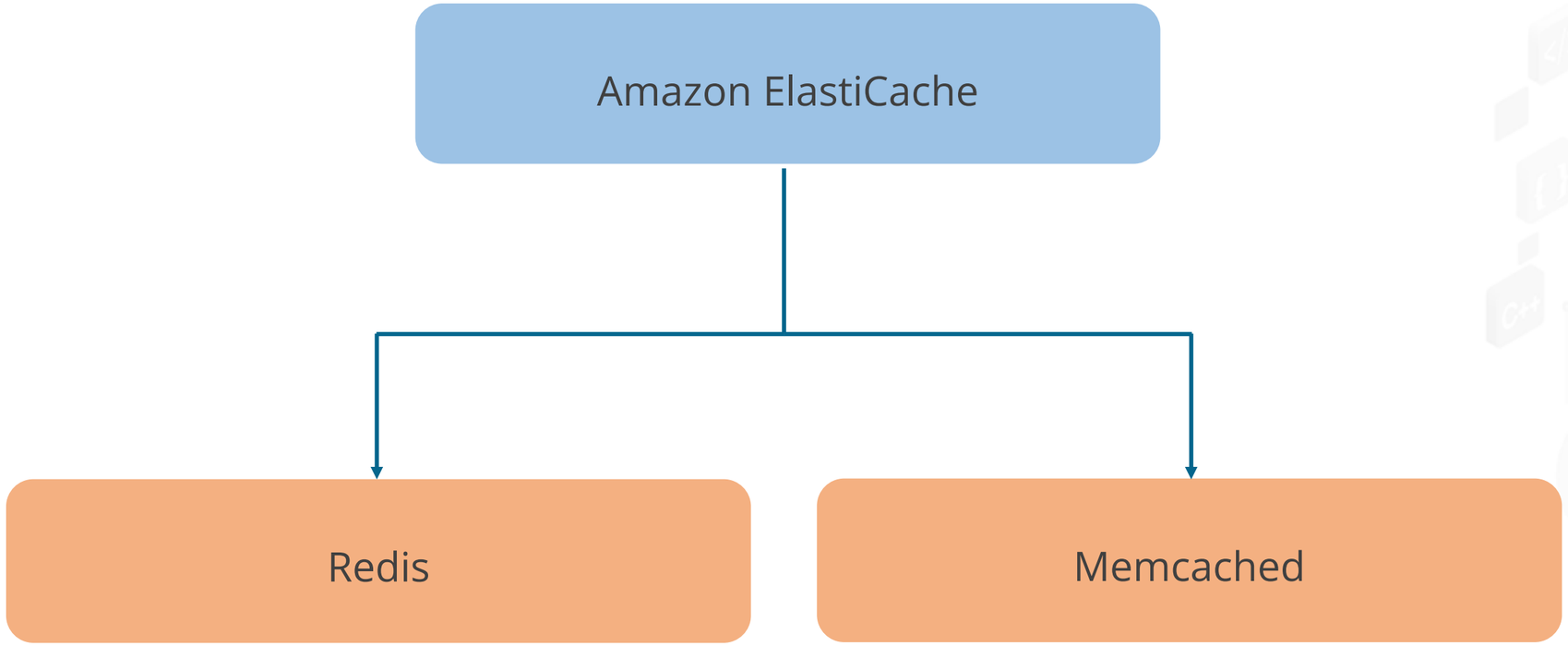
Amazon ElastiCache: Data Engines

Amazon ElastiCache supports two different data engines:



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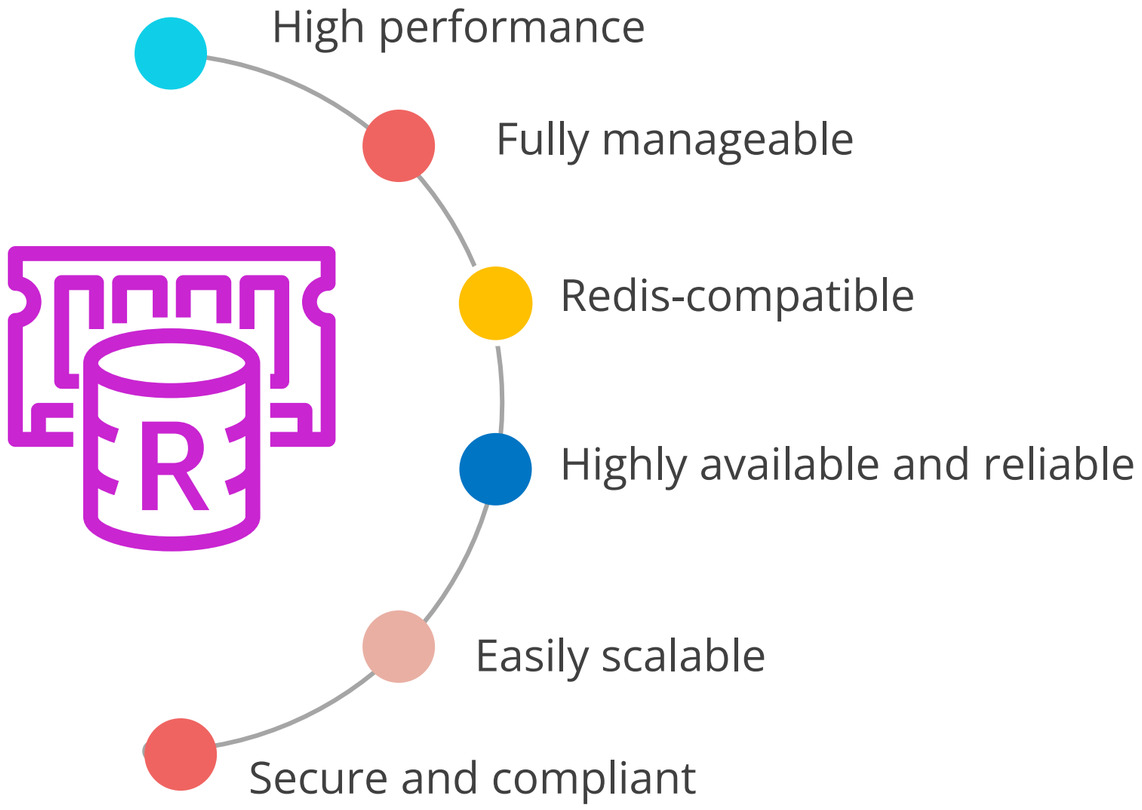
Amazon ElastiCache for Redis

It is a Redis-compatible in-memory data store service and can support high performance since it is fully managed, scalable, and secure. The benefits are as follows:



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Amazon ElastiCache for Memcached

It is a Memcached-compatible in-memory key-value store service that can be used as a cache or a data store. The benefits of Amazon ElastiCache for Memcached are as follows:

- High performance
- High security
- Memcached-compatible
- Easily scalable
- Fully manageable
- Auto discovery

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Quick Check

You are developing an application that experiences high read traffic and requires sub-millisecond response times for frequently accessed data. Which AWS service should you use to cache this data?

A. Amazon RDS

B. Amazon DynamoDB

C. Amazon ElastiCache

D. Amazon S3

Amazon Redshift

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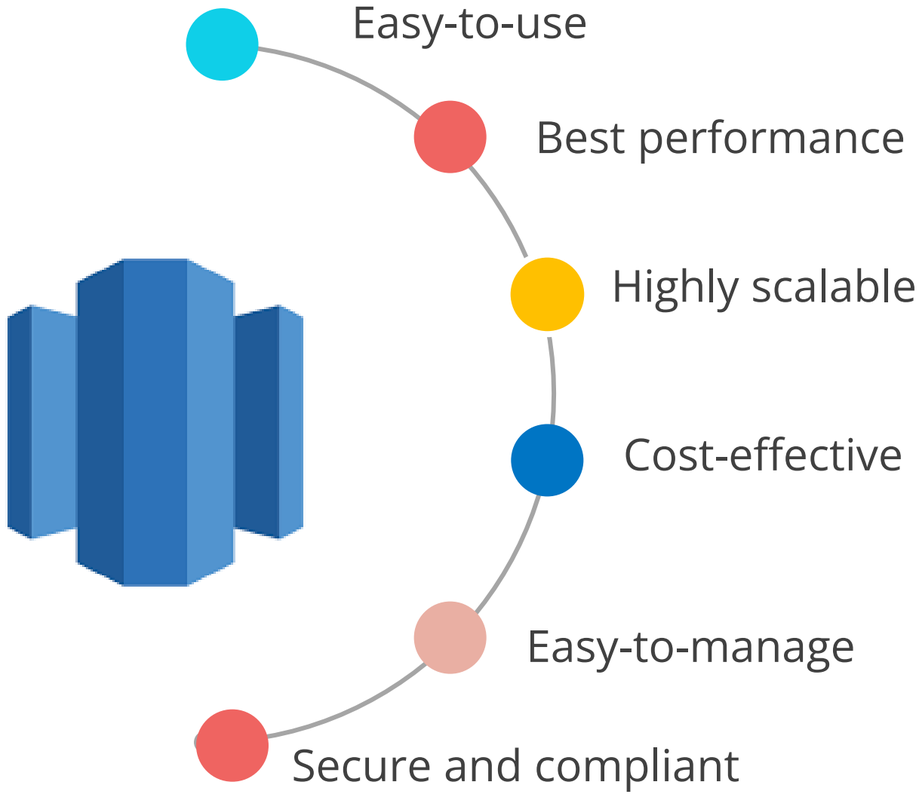
Amazon Redshift

It is a fully managed, petabyte-scale data warehouse service in the cloud.
The benefits of Amazon Redshift are as follows:



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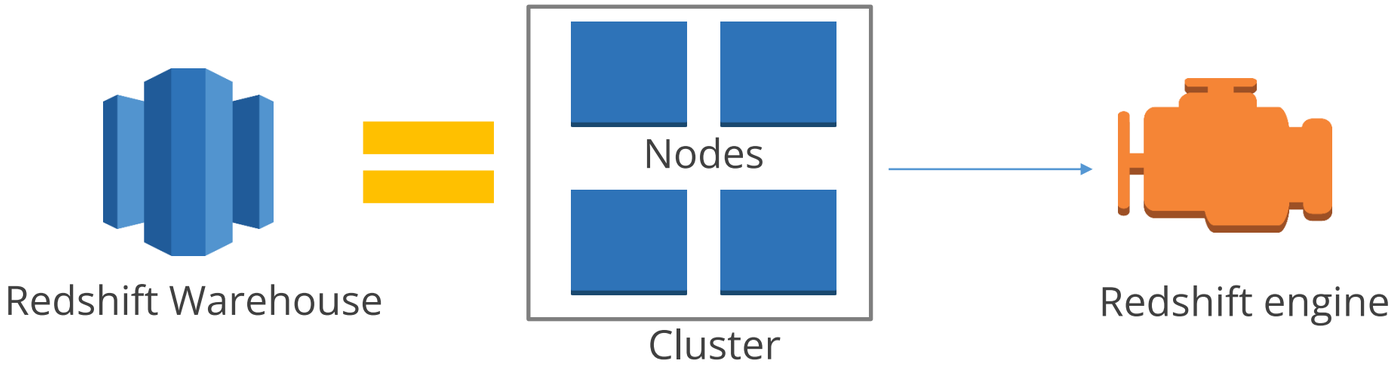
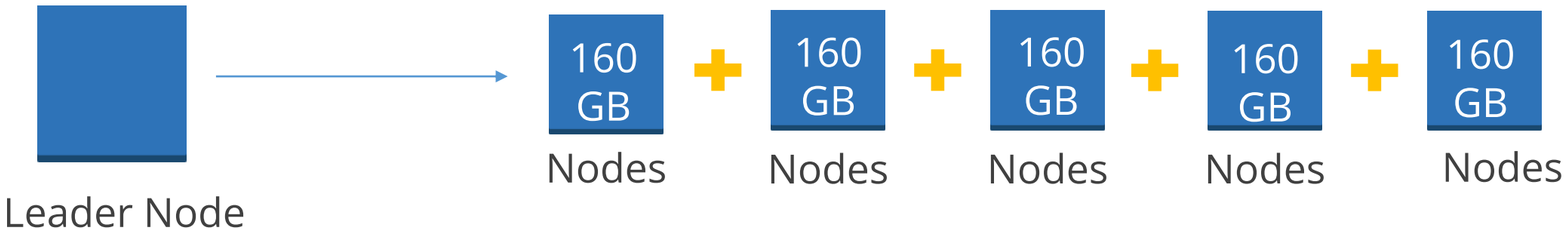
Amazon Redshift Clusters

It is a collection of computing resources called nodes. Nodes are organized into a group called a cluster. Each cluster runs an Amazon Redshift engine and contains one or more databases.





Massive Parallel Processing

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Amazon Redshift Costs

The following are the costs associated with Amazon Redshift:

Backup storage

Managed storage

Data transfer

Compute node hours

Data scan

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Amazon Redshift: Use Cases

The use cases of Amazon Redshift:

- Data analytics as a service
- Session store
- Streaming data analytics
- Real-time analytics

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Configuring Query Data Using Redshift Query Editor

Duration:15 min

Problem Statement:

You have been assigned a task to create a table in Amazon Redshift and access the Redshift Query Editor.

Outcome:

You will be able to create a table in Amazon Redshift, configure it, and perform queries using the Redshift Query Editor.

Note: Refer to the demo document for detailed steps:
03_Configuring_Query_Data_Using_Redshift_Query_Editor

ASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create a table in Redshift

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Creating a RDS MySQL Database

Duration:15 min

Problem Statement:

You have been assigned a task to demonstrate the process of creating an Amazon RDS MySQL database using the AWS Management Console.

Outcome:

You will be able to demonstrate the process of creating an Amazon RDS MySQL database using the AWS Management Console, configure the database settings, manage users, and understand the fundamental tasks required for its operation.

Note: Refer to the demo document for detailed steps:
04_Creating_a_RDS_MYSQL_Database

ASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create a RDS database

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Quick Check

You need to analyze large datasets using SQL-based queries and require a fully managed data warehouse solution that can scale to petabytes of data. Which AWS service should you choose?

A. Amazon RDS

B. Amazon DynamoDB

C. Amazon Redshift

D. Amazon Aurora

Key Takeaways

- Amazon RDS supports several database options including Amazon Aurora, Oracle, Microsoft SQL Server, PostgreSQL, MySQL, and MariaDB.
- Amazon DynamoDB is a NoSQL database service, offering fast, predictable performance and seamless scalability.
- Amazon ElastiCache is a web service that simplifies the deployment, operation, and scaling of an in-memory data store.
- Amazon DocumentDB is designed for managing and scaling JSON formatted data.

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Deploying MySQL RDS Using AWS

Duration: 30 mins

Project agenda: To create and configure an RDS instance

Description: You must create an RDS database and deploy a Linux instance by creating it in EC2 and connecting an SSH client through EC2.

Perform the following:

1. Create an RDS database
2. Launch an EC2 instance
3. Create security groups
4. Connect the terminal to SSH

LESSON-END PROJECT

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Thank You

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